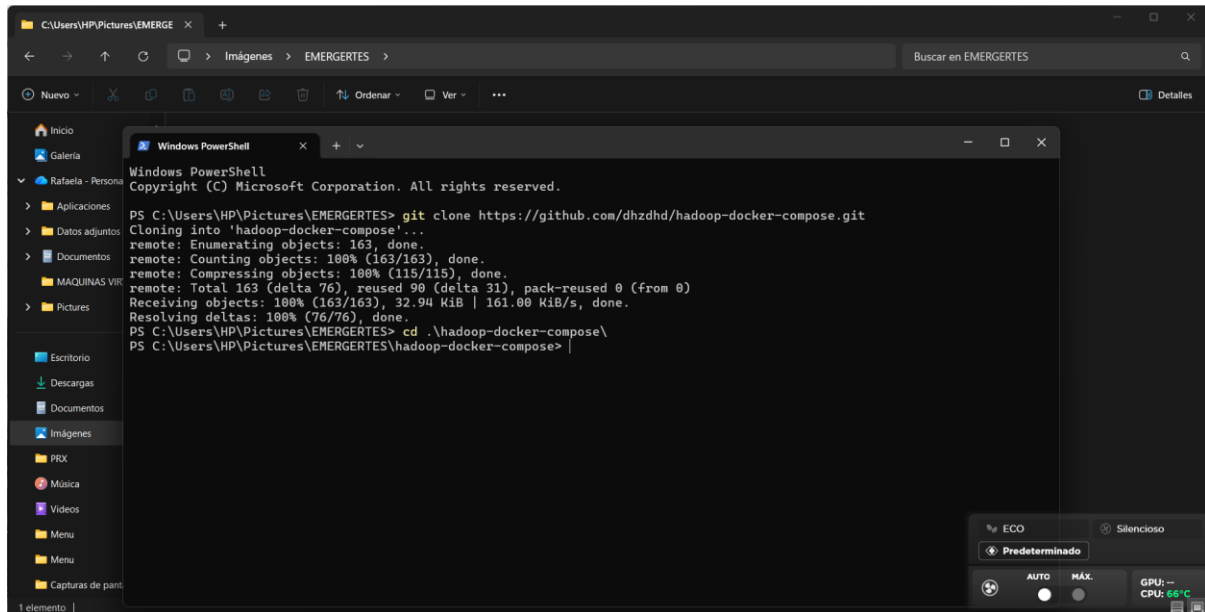


Paso 1

Clonar el repositorio:

git clone URL_DEL_REPOSITORIO

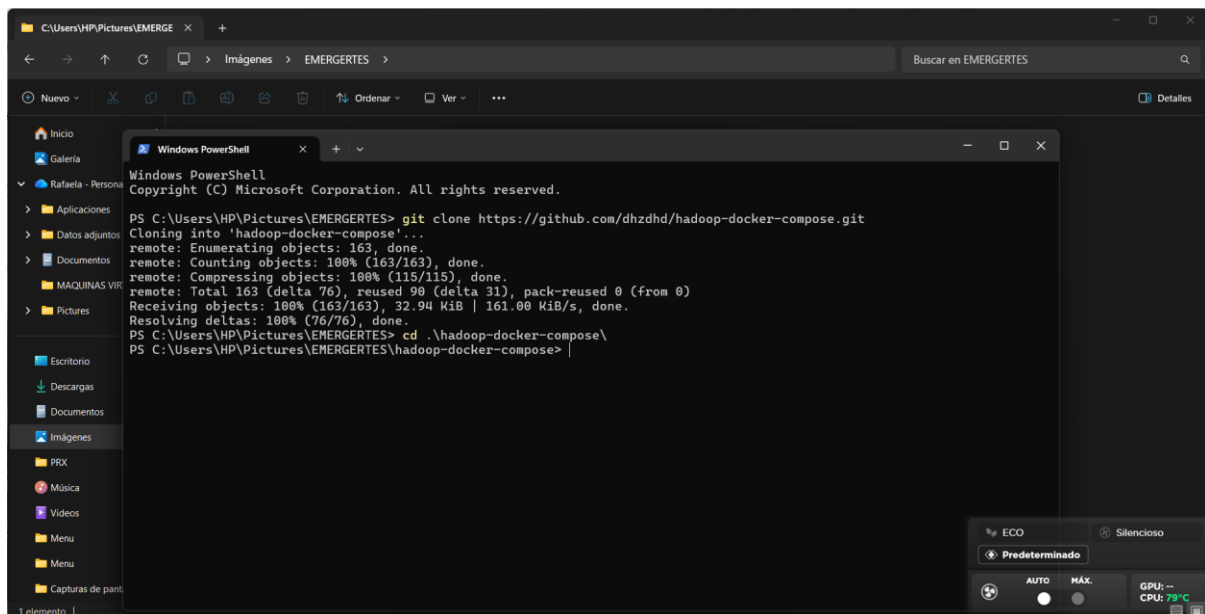
git clone <https://github.com/dhzdhd/hadoop-docker-compose.git>



Paso 2

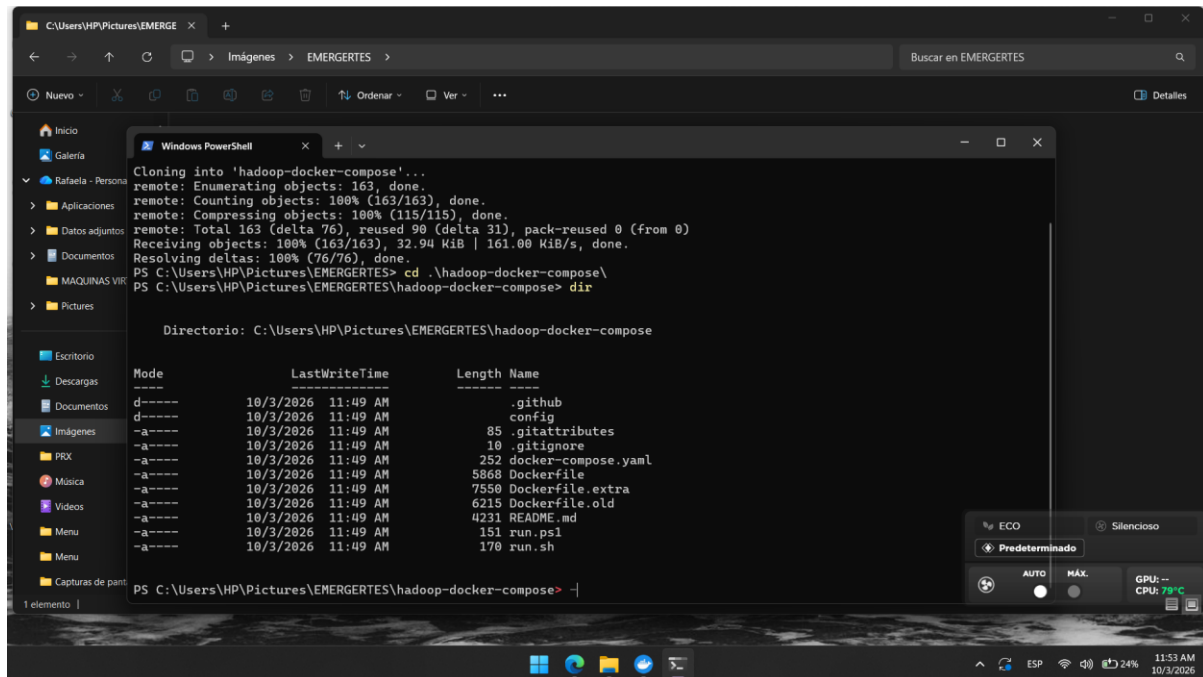
Ingresar al directorio:

cd NOMBRE_DEL_REPOSITORIO



Paso 3

Identificar los archivos principales:
dir



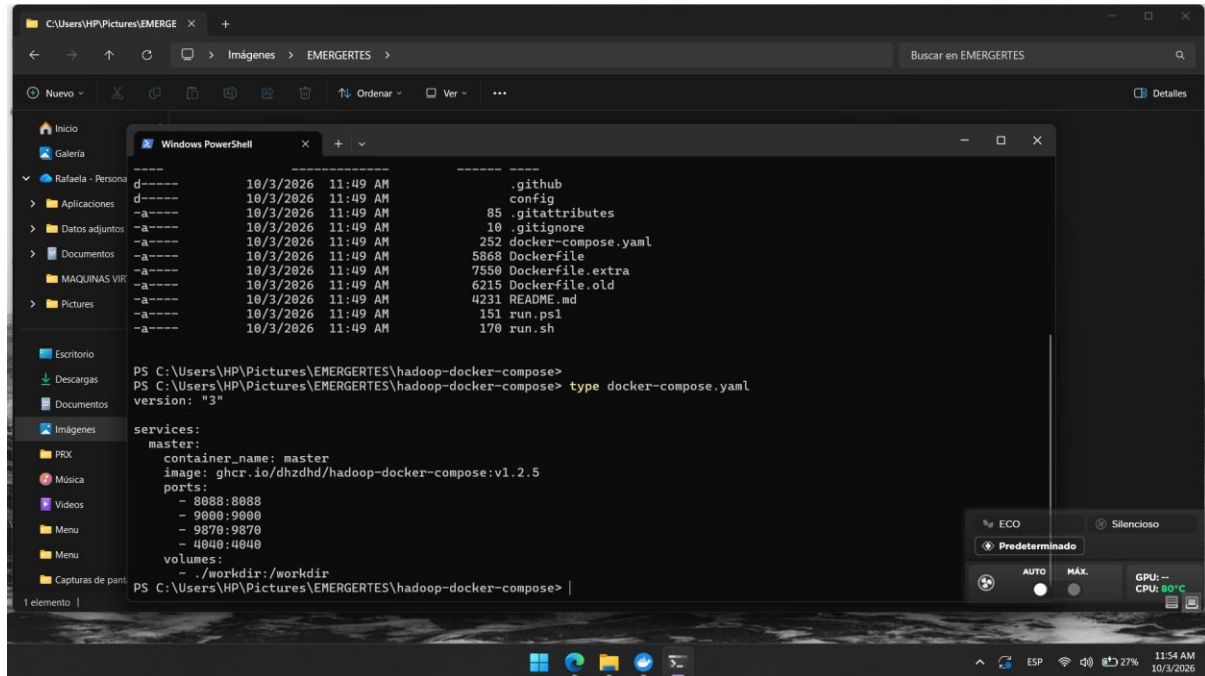
- docker-compose.yaml: define los servicios.
- Dockerfile, Es el principal. Define cómo se construye la imagen que usas (Hadoop, Spark, Hive, Pig, HBase).
- Dockerfile.extra: Variante que agrega Kafka, ZooKeeper y Mahout (el README los asocia a init-extra).
- Dockerfile.old: Versión antigua, queda como respaldo.
- config/: archivos de configuración de Hadoop.
- run.sh / run.ps1: scripts de arranque para Linux y Windows.

Paso 4

Analizar el archivo de Docker Compose:

```
cat docker-compose.yml
```

```
type docker-compose.yml
```

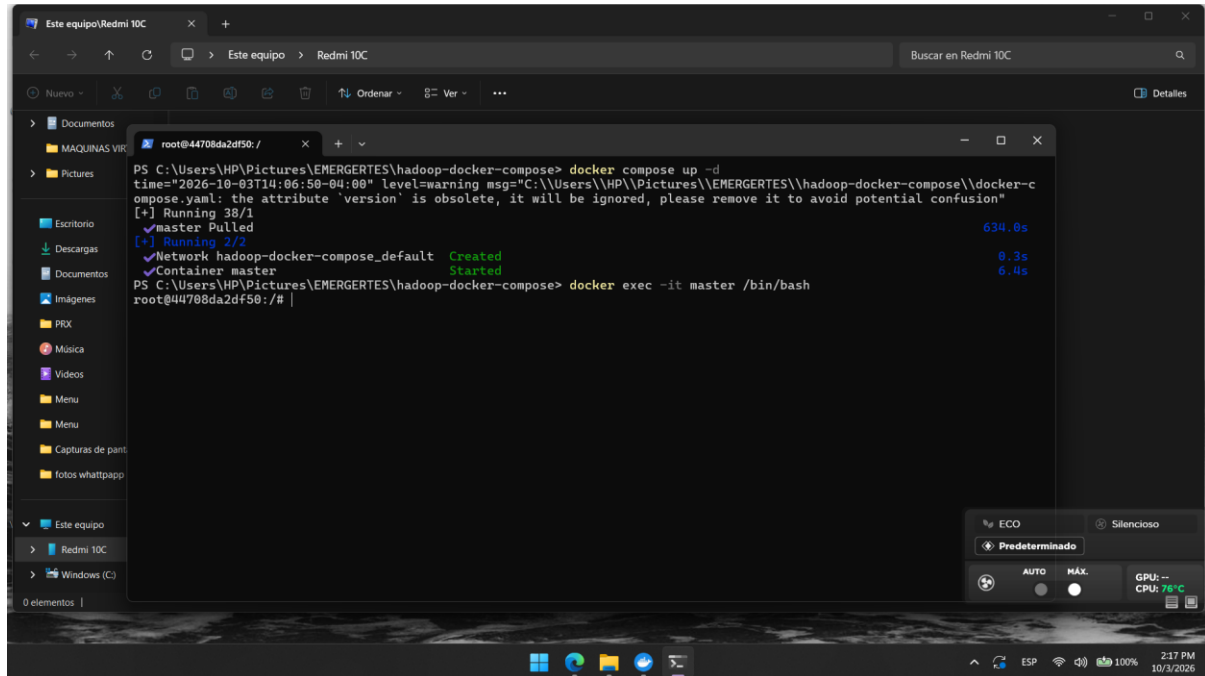


Paso 5

Construir y ejecutar manualmente:

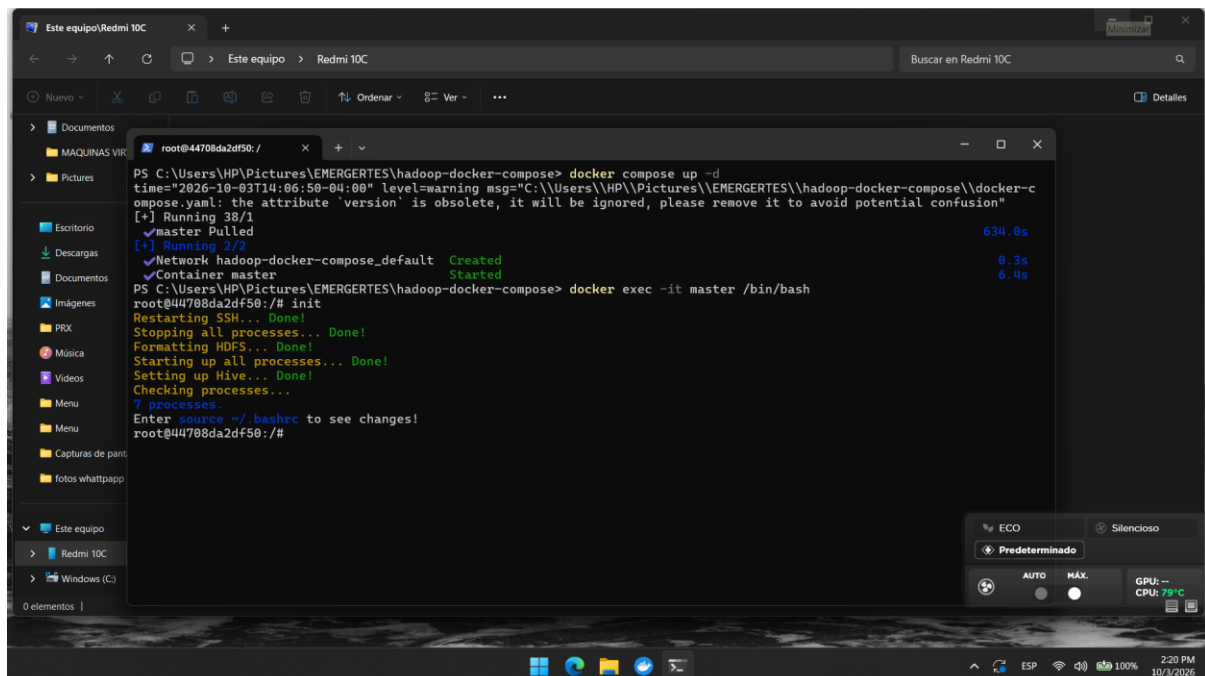
`docker compose up -d`

`docker exec -it master /bin/bash`



```
PS C:\Users\HP\Pictures\EMERGERTES\hadoop-docker-compose> docker compose up -d
time="2026-10-03T14:06:50-04:00" level=warning msg="C:\Users\HP\Pictures\EMERGERTES\hadoop-docker-compose\docker-c
ompose.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 3/1
  ✓ master Pulled                                         634.0s
  [+] Running 2/2
    ✓ Network hadoop-docker-compose_default Created      0.3s
    ✓ Container master Started                          6.4s
PS C:\Users\HP\Pictures\EMERGERTES\hadoop-docker-compose> docker exec -it master /bin/bash
root@44708da2df50:/#
```

Inicializar los contenedores

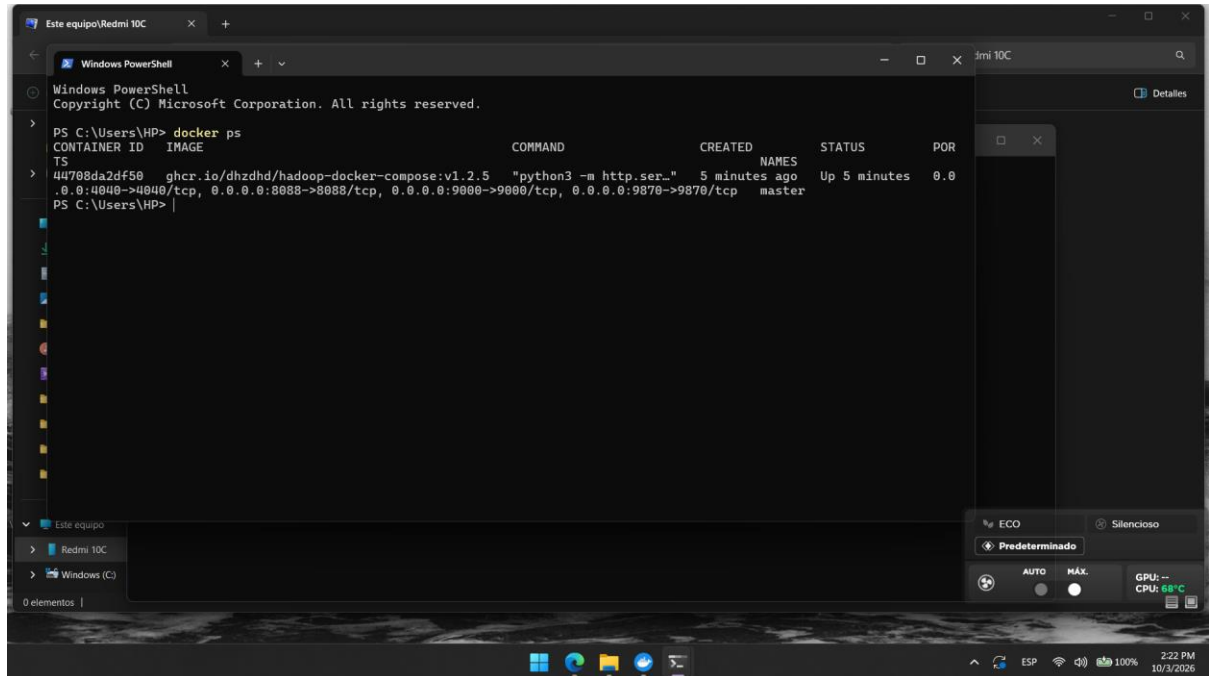


```
PS C:\Users\HP\Pictures\EMERGERTES\hadoop-docker-compose> docker compose up -d
time="2026-10-03T14:06:50-04:00" level=warning msg="C:\Users\HP\Pictures\EMERGERTES\hadoop-docker-compose\docker-c
ompose.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 3/1
  ✓ master Pulled                                         634.0s
  [+] Running 2/2
    ✓ Network hadoop-docker-compose_default Created      0.3s
    ✓ Container master Started                          6.4s
PS C:\Users\HP\Pictures\EMERGERTES\hadoop-docker-compose> docker exec -it master /bin/bash
root@44708da2df50:/# init
Restarting SSH... Done!
Stopping all processes... Done!
Formatting HDFS... Done!
Starting up all processes... Done!
Setting up Hive... Done!
Checking processes...
7 processes.
Enter source ~/.bashrc to see changes!
root@44708da2df50:/#
```

Paso 6

Verificar los contenedores (se hace desde otra terminal):

docker ps



Paso 7

Realizar una prueba funcional relacionada con la tecnología seleccionada.

1. Verificar que HDFS responde

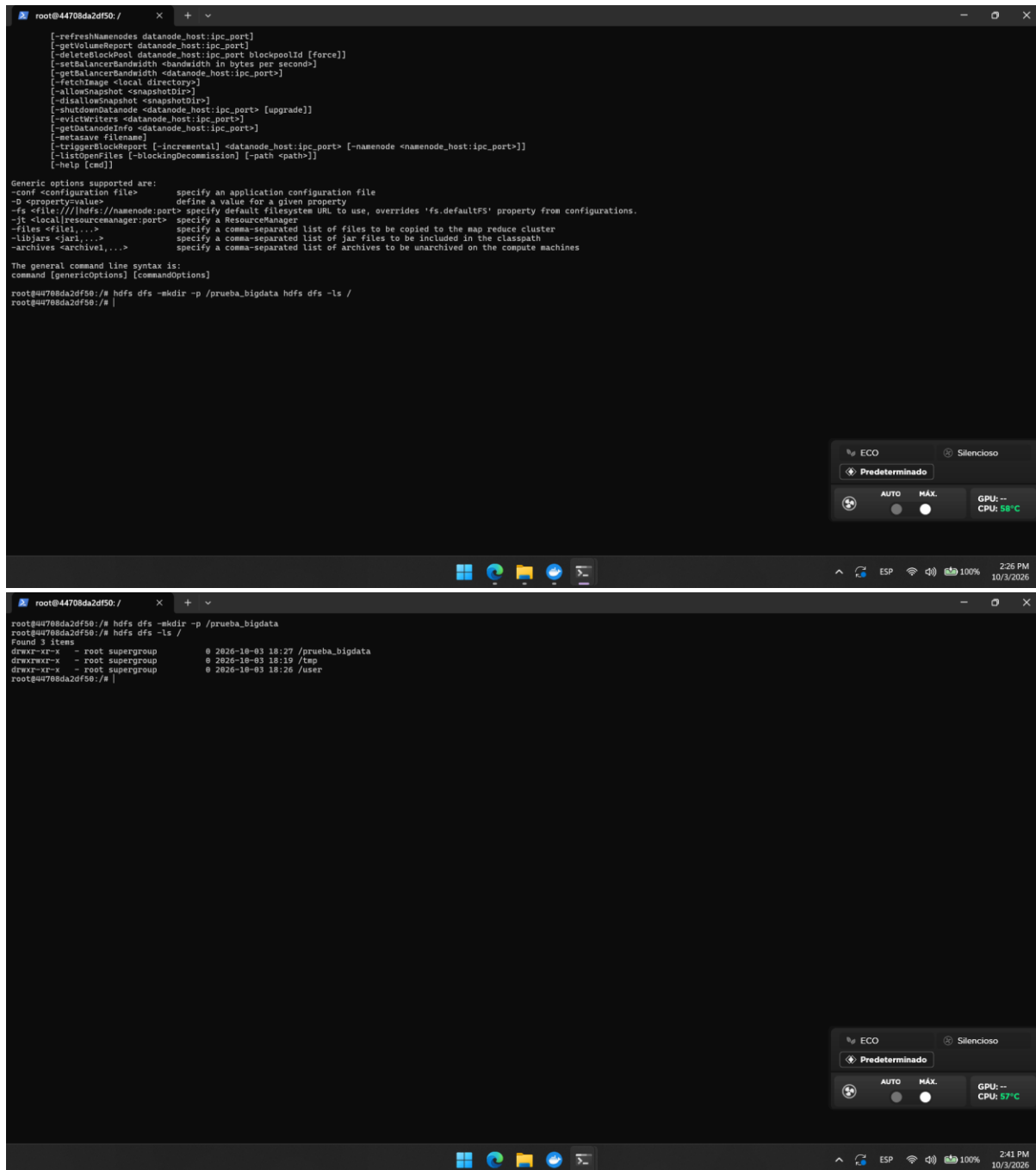
hdfs dfsadmin -report

```
root@44708da2df50: /  
root@44708da2df50:/# hdfs dfsadmin -report  
Configured Capacity: 1081181176832 (1006.85 GB)  
Present Capacity: 1080429776896 (939.17 GB)  
DFS Remaining: 1080429771840 (939.17 GB)  
DFS Used: 45856 (44 KB)  
DFS Used%: 0.00%  
Replicated Blocks:  
  Under replicated blocks: 1  
  Blocks with corrupt replicas: 0  
  Missing blocks: 0  
  Missing blocks (with replication factor 1): 0  
  Low redundancy blocks with highest priority to recover: 1  
  Pending deletion blocks: 0  
Erasure Coded Block Groups:  
  Low redundancy block groups: 0  
  Block groups with corrupt internal blocks: 0  
  Missing block groups: 0  
  Low redundancy blocks with highest priority to recover: 0  
  Pending deletion blocks: 0  
-----  
Live datanodes (1):  
Name: 172.21.0.2:9866 (44708da2df50)  
Hostname: 44708da2df50  
Decommission Status : Normal  
Configured Capacity: 1081181176832 (1006.85 GB)  
DFS Used: 45856 (44 KB)  
Non DFS Used: 17679044608 (16.46 GB)  
DFS Remaining: 1080429771840 (939.17 GB)  
DFS Used%: 0.00%  
DFS Remaining%: 93.28%  
Configured Cache Capacity: 0 (0 B)  
Cache Used: 0 (0 B)  
Cache Remaining: 0 (0 B)  
Cache Used%: 100.00%  
Cache Remaining%: 0.00%  
Xceivers: 0  
Last contact: Sat Oct 03 18:39:52 GMT 2026  
Last Block Report: Sat Oct 03 18:19:04 GMT 2026  
Num of Blocks: 1  
  
root@44708da2df50:/# |
```

2. Crear un directorio

hdfs dfs -mkdir -p /prueba_bigdata

hdfs dfs -ls /



The image shows two screenshots of a terminal window. The top screenshot displays the help text for the 'hdfs dfs' command, listing various subcommands like [-refreshNameNodes], [-getVolumeReport], [-deleteBlockPool], [-setBalancerBandwidth], [-getBalancerBandwidth], [-fetchImage], [-allowSnapshot], [-disallowSnapshot], [-shutdownDatanode], [-evictWriters], [-getDatanodeInfo], [-metasave], [-triggerBlockReport], [-listOpenFiles], and [-help]. It also lists generic options supported: [-conf], [-D], [-fs], [-jt], [-files], [-libjars], and [-archives]. The bottom screenshot shows the execution of the 'hdfs dfs -mkdir -p /prueba_bigdata' command, followed by 'hdfs dfs -ls /', which lists the contents of the root directory, showing three items: 'drwxr-xr-x - root supergroup 0 2026-10-03 18:27 /prueba_bigdata', 'drwxr-xr-x - root supergroup 0 2026-10-03 18:15 /tmp', and 'drwxr-xr-x - root supergroup 0 2026-10-03 18:26 /user'.

```
root@44708da2df50: /
[ -refreshNameNodes datanode_host:ipc_port]
[ -getVolumeReport datanode_host:ipc_port]
[ -deleteBlockPool datanode_host:ipc_port blockpoolid [force]]
[ -setBalancerBandwidth <bandwidth in bytes per second>]
[ -getBalancerBandwidth <datanode_host:ipc_port>]
[ -fetchImage <local directory>]
[ -allowSnapshot <snapshotDir>]
[ -disallowSnapshot <snapshotDir>]
[ -shutdownDatanode <datanode_host:ipc_port> [upgrade]]
[ -evictWriters <datanode_host:ipc_port>]
[ -getDatanodeInfo <datanode_host:ipc_port>]
[ -metasave filename]
[ -triggerBlockReport [-incremental] <datanode_host:ipc_port> [-namenode <namenode_host:ipc_port>]]
[ -listOpenFiles [-blockingDecommission] [-path <path>]]
[ -help [cmd]]

Generic options supported are:
-conf <configuration file>      specify an application configuration file
-D <property=value>             define a value for a given property
-fs <file:///hdfs://namenode:port> specify default filesystem URL to use, overrides 'fs.defaultFS' property from configurations.
-jt <localresourceemanager:port> specify a ResourceManager
-files <file1,...>               specify a comma-separated list of files to be copied to the map reduce cluster
-libjars <jar1,...>             specify a comma-separated list of jar files to be included in the classpath
-archives <archive1,...>       specify a comma-separated list of archives to be unarchived on the compute machines

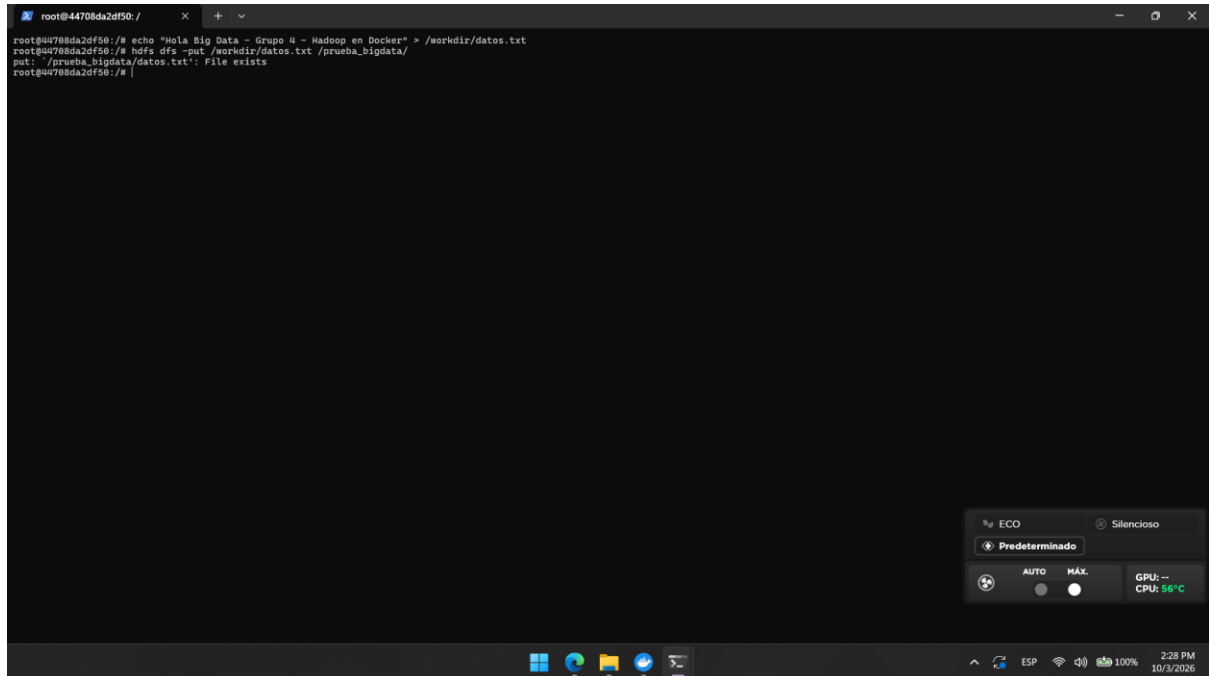
The general command line syntax is:
command [genericOptions] [commandOptions]

root@44708da2df50: /# hdfs dfs -mkdir -p /prueba_bigdata hdfs dfs -ls /
root@44708da2df50: /#
root@44708da2df50: /# hdfs dfs -mkdir -p /prueba_bigdata
root@44708da2df50: /# hdfs dfs -ls /
Found 3 items
drwxr-xr-x - root supergroup 0 2026-10-03 18:27 /prueba_bigdata
drwxr-xr-x - root supergroup 0 2026-10-03 18:15 /tmp
drwxr-xr-x - root supergroup 0 2026-10-03 18:26 /user
root@44708da2df50: /#
```

3. Crear un archivo local y cargarlo

```
echo "Hola Big Data - Grupo X - Hadoop en Docker" > /workdir/datos.txt
```

```
hdfs dfs -put /workdir/datos.txt /prueba_bigdata/
```

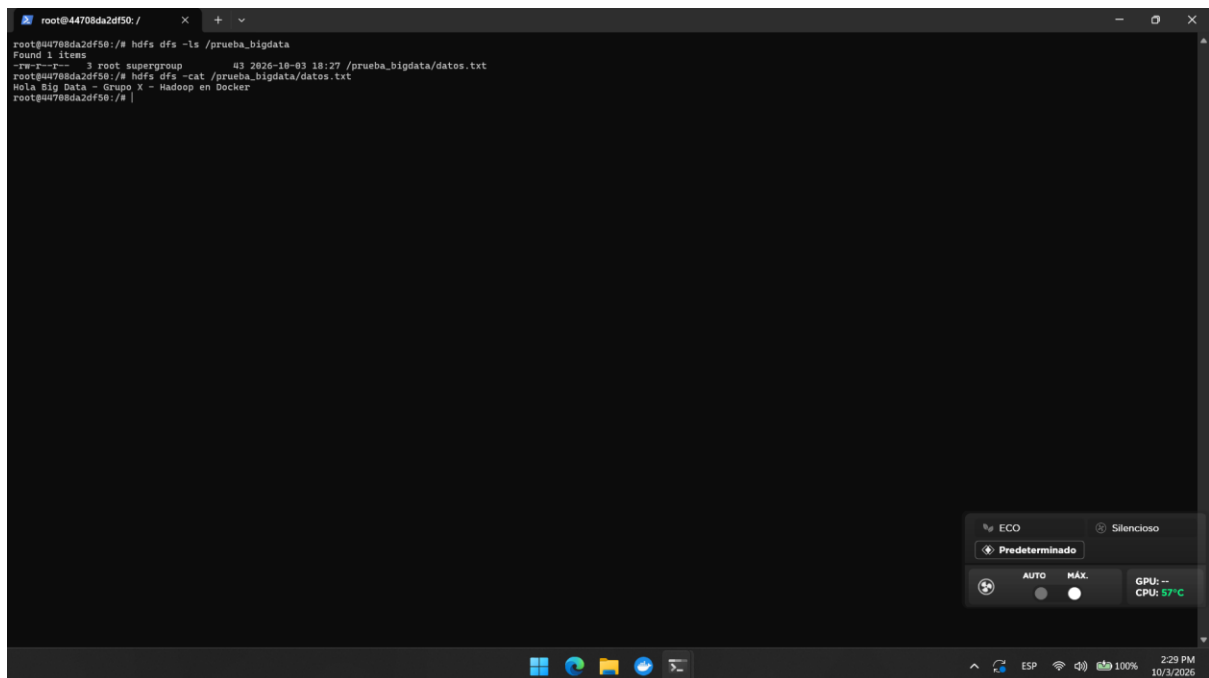


A terminal window with a dark background. The prompt is `root@44708da2df50:/`. The user enters `echo "Hola Big Data - Grupo X - Hadoop en Docker" > /workdir/datos.txt`. The prompt changes to `root@44708da2df50:/` and the user enters `hdfs dfs -put /workdir/datos.txt /prueba_bigdata/`. The output is `put: /prueba_bigdata/datos.txt: File exists`. The prompt returns to `root@44708da2df50:/`. In the bottom right corner, there is a system tray with a fan icon, a temperature gauge showing `GPU: -- CPU: 56°C`, and a status bar showing `2:28 PM 10/3/2026`.

4. Consultar el archivo almacenado

```
hdfs dfs -ls /prueba_bigdata
```

```
hdfs dfs -cat /prueba_bigdata/datos.txt
```



A terminal window with a dark background. The prompt is `root@44708da2df50:/`. The user enters `hdfs dfs -ls /prueba_bigdata`. The output is `Found 1 items`. The prompt changes to `root@44708da2df50:/` and the user enters `hdfs dfs -cat /prueba_bigdata/datos.txt`. The output is `Hola Big Data - Grupo X - Hadoop en Docker`. The prompt returns to `root@44708da2df50:/`. In the bottom right corner, there is a system tray with a fan icon, a temperature gauge showing `GPU: -- CPU: 57°C`, and a status bar showing `2:29 PM 10/3/2026`.

